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PG 1

① GIVEN: $\Delta T_H: 10 \times 10^{-6}$ $\Delta p_o: 0.015$
 $F_R: 8 \times 10^3 \text{ fiss/cm}^3 \cdot \text{s}$ $B_D = 5 \text{ MW/D/kgU}$
 $T_F = 1500 \text{ K}$ $p_{O_2}: 10.97 \text{ g/cc}$
 $T_{ref} = 300 \text{ K}$ $t: 300 \text{ DAYS}$ UNITS

$\beta = \frac{F_R t}{N_U}$, $E_{SFP} = 5.577 \times 10^{-2} p \beta$

$E_{GFP} = 1.96 \times 10^{-28} p \beta (2800 - T)^{11.73} e^{-0.0162(2800 - T)} e^{-17.8 p \beta}$

$t_{sec} = \frac{300 \text{ DAYS} | 24 \text{ HRS} | 3600 \text{ SEC}}{1 \text{ DAY} | 1 \text{ HRS}} = 2.59 \times 10^7 \text{ s}$

NEED N_U :

ASSUME A MASS $UO_2 \approx 270 \text{ g/mol}$ TO

$N_U = \frac{P}{M_{UO_2}} N_A = \frac{10.97 \text{ g/cc}}{270 \text{ g/mol}} \cdot 6.022 \times 10^{23}$

$N_U = 2.45 \times 10^{22} \text{ U/kms/cc}$

$\beta = \frac{F_R t}{N_U} = \frac{8 \times 10^3 \cdot 2.59 \times 10^7 \text{ s}}{2.45 \times 10^{22}} = 0.0846$

ASSUME AN ISOTROPIC SOLID!

$E_{TH} = 3 \Delta T_H (T_F - T_{ref})$
 $= 3 \cdot (10 \times 10^{-6}) (1500 \text{ K} - 300 \text{ K}) = 0.036$

not needing the 3

$E_{SFP} = 5.577 \times 10^{-2} p \beta$
 $= 5.577 \times 10^{-2} (10.97) (0.0846)$
 $= 0.0518$

$E_{GFP} = 1.96 \times 10^{-28} p \beta (2800 - T)^{11.73} e^{-0.0162(2800 - T)} e^{-17.8 p \beta}$
 $= 1.96 \times 10^{-28} (10.97) (0.0846) (2800 - 1500)^{11.73} e^{-0.0162(2800 - 1500)} e^{-17.8 (10.97) (0.0846)}$
 $= 2.9 \times 10^{-8}$

ASSUME $E_D = \Delta p_o = 0.015$

$\Delta V/V = E_{TH} + E_{SFP} + E_{GFP} + E_D = 0.036 + 0.0518 + 2.9 \times 10^{-8} + (-0.015)$
 $= 0.0728 \text{ OR } 7.28\%$

Pg 2

② GIVEN: $A_0 = 3.14 \times 10^{24}$

$$G = 4.2519 \times 10^{10} - 2.2185 \times 10^7 T$$

$$n = 5$$

$$Q = 2.7 \times 10^5$$

$$C_0 = 1.654 \times 10^{-24}$$

$$C_1 = 0.8 \text{ S}$$

$$C_2 = 1$$

$$\text{LHR} = 250 \text{ W/cm}$$

$$T = 600 \text{ K}$$

$$t = 200 \text{ DAYS}$$

$$\sigma = 60 \text{ MPa}$$

$$\begin{cases} \dot{\epsilon}_{TH} = A_0 \left(\frac{\sigma}{G} \right)^n e^{-Q/RT} \\ \dot{\epsilon}_{LR} = C_0 \phi^{C_1} \sigma^{C_2} \\ \phi = 3 \times 10^{-11} \times \text{LHR} \end{cases}$$

$$G = 4.2519 \times 10^{10} - 2.2185 \times 10^7 (600)$$

$$G = 2.92 \times 10^{10} \text{ Pa}$$

$$\dot{\epsilon}_{TH} = A_0 \left(\frac{\sigma}{G} \right)^n e^{-Q/RT}$$

$$\dot{\epsilon}_{TH} = 3.14 \times 10^{24} \left(\frac{60 \times 10^6 \text{ Pa}}{2.92 \times 10^{10} \text{ Pa}} \right)^5 \cdot e^{(-2.7 \times 10^5 / (8.314 \cdot 600))}$$

$$\dot{\epsilon}_{TH} = 3.583 \times 10^{-13}$$

$$\phi = 3 \times 10^{-11} \cdot \text{LHR} = 3 \times 10^{-11} (250) = 7.5 \times 10^{-9}$$

$$\begin{aligned} \dot{\epsilon}_{LR} &= C_0 \phi^{C_1} \sigma^{C_2} \\ &= 1.654 \times 10^{-24} (7.5 \times 10^{-9})^{0.85} (60)^1 \\ &= 6.17 \times 10^{-11} \end{aligned}$$

$$\dot{\epsilon}_{TOT} = \dot{\epsilon}_{TH} + \dot{\epsilon}_{LR} = 3.583 \times 10^{-13} + 6.17 \times 10^{-11}$$

$$\dot{\epsilon}_{TOT} = 6.21 \times 10^{-11}$$

For 200 DAYS:

$$200 \text{ DAYS} \left| \frac{24 \text{ HR}}{\text{DAY}} \right| \frac{3600 \text{ S}}{\text{HR}} = 1.73 \times 10^7 \text{ S} = t$$

$$\epsilon_{TOT}(t) = 6.21 \times 10^{-11} (1.73 \times 10^7 \text{ S})$$

$$\epsilon_{TOT}(200 \text{ DAYS}) = \boxed{1.07 \times 10^{-3} \text{ or } 0.107\%}$$

P63

(3) GIVEN: $t = 250 \text{ DAYS}$

$$T = +51400\text{K}$$

$$\dot{\Gamma} = 2 \times 10^{13} \text{ fss/cm}^3 \cdot \text{s}$$

GRAIN SIZES: $5\mu\text{m}$ $40\mu\text{m}$

$$k_B = 8.617 \times 10^{-5} \text{ eV/K}$$

$$D_T = D_1 + D_2 + D_3 \text{ cm}^2/\text{s}$$

$$D_1 = 7.6 \times 10^{-6} \exp\left(\frac{-3.03}{k_B T}\right)$$

$$\tau = \frac{D_T}{a^2}$$

$$D_2 = 1.41 \times 10^{-18} \exp\left(\frac{-1.19}{k_B T}\right) \sqrt{\dot{\Gamma}}$$

$$D_3 = 2.0 \times 10^{-30} \cdot \dot{\Gamma}$$

$$D_1 = 7.6 \times 10^{-6} \exp\left(\frac{-3.03}{8.617 \times 10^{-5} \cdot 51400}\right) = 9.41 \times 10^{-17} \text{ cm}^2/\text{s}$$

$$D_2 = 1.41 \times 10^{-18} \exp\left(\frac{-1.19}{8.617 \times 10^{-5} (51400)}\right) = 3.28 \times 10^{-16} \text{ cm}^2/\text{s}$$

$$D_3 = 2.0 \times 10^{-30} \cdot \dot{\Gamma} = 2.0 \times 10^{-30} (2 \times 10^{13}) = 4 \times 10^{-17} \text{ cm}^2/\text{s}$$

$$D_T = 9.41 \times 10^{-17} + 3.28 \times 10^{-16} + 4 \times 10^{-17} = 4.62 \times 10^{-16} \text{ cm}^2/\text{s}$$

$$\frac{t = 250 \text{ DAYS}}{250} \left| \begin{array}{c} 24 \text{ hr} \\ 3600 \text{ s} \end{array} \right| = \frac{2.16 \times 10^7 \text{ sec}}{1.78173 \times 10^7 \text{ sec}}$$

$$\frac{5\mu\text{m GRAIN}}{\tau = \frac{D_T}{a^2}} \quad \text{ASSUME } a = \text{GRAIN LENGTH} = 5 \times 10^{-4} \text{ cm}$$

$$\tau = \frac{4.62 \times 10^{-16} \text{ cm}^2/\text{s} (2.16 \times 10^7 \text{ sec})}{(5 \times 10^{-4} \text{ cm})^2}$$

$$\tau = 0.0399$$

$$\pi^2 = 0.101 \Rightarrow \tau < \pi^2 \Rightarrow f = 4 \sqrt{\frac{\tau}{\pi a^2}} - \frac{3}{2} \frac{\tau}{a^2}$$

$$f = 4 \sqrt{\frac{0.0399}{\pi (5 \times 10^{-4})^2}} - \frac{3}{2} (0.0399)$$

$$f = 0.391 \text{ or } 39.1\% \quad \text{FOR } 5\mu\text{m GRAIN}$$

$$\frac{40\mu\text{m GRAIN}}{\tau = \frac{4.62 \times 10^{-16} (2.16 \times 10^7)}{(4 \times 10^{-3})^2}} = 6.24 \times 10^{-4} < \pi^2$$

$$f = 4 \sqrt{\frac{6.24 \times 10^{-4}}{\pi}} - \frac{3}{2} (6.24 \times 10^{-4}) = 0.0553 \text{ or } 5.53\% \quad \text{FOR } 40\mu\text{m GRAIN}$$

PG 4

4/4
④ IN UO_2 THE TYPICAL CHARGE IS U^{4+} . IT CAN VARY BASED ON OXYGEN-METAL RATIO DUE TO OTHER CHARGE VARIANCES IN U . SUCH AS U^{3+} , U^{5+} , U^{6+} .

4/5
⑤ DURING RX OPERATION, FISSION PRODUCTS REACT W/ O_2 AND THE CLADDING AND CAN CHANGE DUE TO THE O_2 /METAL RATIO. MO IS A BUFFERING AGENT TO MAINTAIN O_2 LOWER NEAR CLADDING EDGES + MAINTAIN O_2 RATIOS IN FUEL PELLETS. THIS CAN HELP CONTROL FISSION PRODUCT PHASES.

3/3
⑥ MELTING TEMP, THERMAL CONDUCTIVITY, CREEP
- mostly

8/10
⑦ SOLUBLE OXIDE - LOWERS THERMAL CONDUCTIVITY AS A PHONON SCATTERER

INSOLUBLE OXIDE - ALTERS MICROSTRUCTURE + THERMAL BEHAVIOR

METALS - MODIFY MICROSTRUCTURE

VOLATILES - CAUSE CLADDING CORROSION (INTERNAL) (I_2)

NOBLE GAS - $\downarrow$ GAP CONDUCTANCE HE/XE MIX LESS COND.

THEN PUZE HE

- wanted more specifics

8/8
⑧ PART 1 - FISSION GENERATES XE/KR WHICH DIFFUSE TO GRAIN BOUNDARIES.

PART 2 - BUBBLES FORM ON GRAIN BOUNDARIES AND GROW TOGETHER

PART 3 - GAS TRAVELS CONNECTED BOUNDARY BUBBLES TO FREE SURFACE + ENTERS GAP REGION

5/5
⑨ CREEP IS A PLASTIC DEFORMATION OVER TIME FROM ASSOCIATED STRESSORS. ONE MECHANISM IS THE DIFFUSION OF ATOMS OR VACANCIES ALONG GRAIN BOUNDARY LINES KNOWN AS CREEP

PG 5

- 8/8
- ⑩ HBS IS WHEN FUEL RESTRUCTURES ALLOWING IRRADIATION DAMAGE + ENERGY BUILDUP AROUND THE FUEL RIM AREA. THIS CAUSES UNSTABLE CRYSTAL STRUCTURE IN UO_2 CERAMIC.

THIS CAUSES A DENSE POROUS STRUCTURE AND CHANGES THE GRAIN STRUCTURE TO A SMALLER STRUCTURE GRAIN. ~~THIS WILL CAUSE HIGH EMISSION~~ ~~OF NEUTRONS~~ ~~AND~~ ~~THIS WILL DRIVE LOWER~~ PHONON SCATTERING $\Rightarrow$ THERMAL CONDUCTIVITY $\uparrow$, AS POROSITY ALSO $\uparrow$, THE CERAMIC IS SOFTER $\downarrow$ ST.

- 3/6
- ⑪ GROWTH OCCURS BECAUSE INTERSTITIALS + VACANCIES ARE ALIGNED AND CAUSE EXPANSION IN ONE DIRECTION, BUT CONTRACTION IN THE OTHER. WITH MORE NEUTRON IRRADIATION, MORE DEFECTS OCCUR $\Rightarrow$ THE LATTICE STRUCTURE DRIVES OVERALL GROWTH. - more specificity

- 5/8
- ⑫ CLADDING UNDERGOES THERMAL + IRRADIATIVE CREEP WHICH CHANGES ZR CLADDING DIMENSIONS DUE TO CONSTANT PRESSURE EXERTED EXTERNALLY. AS FUEL EXPANDS + SWELLS OVER TIME, THE EFFECTS OF CREEP ASSIST IN MINIMIZING STRESS ON THE CLADDING.

- needed much more

- 10/12
- ⑬ ① CORROSIVE ENVIRONMENT - VOLATILE FISSION PRODUCTS IN THE GAP ATTACK ZR IF ZrO_2 IS BREACHED

② SUSCEPTABLE MATERIAL - ZR IS THE MATERIAL THAT IS SUSCEPTABLE.

③ SUFFICIENT STRESS - PCI ADDS STRESS FROM FUEL/CLADDING INTERACTIONS $\uparrow$ TENSILE HOOP STRESS

④ SUFFICIENT TIME - A FLAW IN OXIDE LAYER MUST FORM OVER INCUBATION TIME.

- needed more everywhere

- 5/5
- ⑭ IODINE REACTS WITH ZR TO FORM ZrI_4 WHICH REACTS WITH O_2 TO FORM ZrO_2 + I_2 WHICH THEN ATTACKS ZR AGAIN. THIS $\uparrow$ CRACK GROWTH WHEN STRESS IS PRESENT.